MOHAMMAD ROHAAN

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OBJECTIVE

Motivated **7th-semester BS Computer Science** student at **FAST National University, Islamabad,** with strong foundations in software engineering, operating systems, embedded systems, and algorithmic problem solving. Experienced in building real-world systems spanning .NET applications, Java desktop software, concurrent systems, enterprise networks, and large-scale algorithmic case studies. Actively exploring AI, machine learning, robotics, and high-performance computing, with hands-on work in Merkle Trees, heap optimizations, and HPCG benchmark analysis. Seeking a software / systems / AI-focused internship to contribute to impactful engineering solutions.

EDUCATION

FAST University Islamabad - Bachelor of Science in Computer Science

7th Semester (Expected Graduation: January 2027)

- Relevant Coursework: Programming Fundamentals (C++), Object-Oriented Programming (C++), Data Structures & Algorithms (C++), Database Management Systems (SQL Server), .NET Framework (C#), Robotics Technology (MATLAB, Kinematics) ,Design & Analysis of Algorithms, Numerical Computing, OS, Computer Networks, Embedded Systems

- **Note:** One-year academic pause due to personal circumstances, resumed with strong academic performance.

EXPERIENCE & ACHIEVEMENTS

- Optimized and analyzed an e-commerce database with over 1 million records using SQL Server, writing complex T-SQL queries and generating ETL reports for business insights.
- Completed a Robotics Technology course, mastering forward/inverse kinematics and developing MATLAB simulations for robotic arm motion planning.

SELECTED PROJECTS

HPCG Kernel Optimization (Numerical Computing)
Optimized the HPCG benchmark by improving SpMV/SymGS kernel performance. Integrated optimized routine into official HPCG codebase, benchmarked results, analyzed memory behavior, and reported kernel-level performance improvements.

Large-Scale Integrity Verification using Merkle Trees (DAA)
Implemented a Merkle Tree system from scratch on 1M+ Amazon reviews. Built existence proofs, tamper detection, and benchmarking modules measuring hash time, proof latency, and memory usage.

Cisco Enterprise Network Simulation
Designed a multi-block enterprise topology using VLSM, OSPF (Area 1 & 2), EIGRP, RIP, inter-protocol redistribution, DHCP, NAT, and ACL-based security policies.

Airline Management System (Operating Systems)
Built a concurrent airline booking simulator in C using pthreads, semaphores, and mutex locks with deadlock prevention and memory safety mechanisms.

Inventory & Stock Management System (Java / SQL)
Developed a JavaFX desktop system with secure login, real-time stock updates, supplier management, low-stock alerts, and automated reporting.

Git Lite (C++, DSA)
Designed a lightweight version control system implementing trees, graphs, diff and merge algorithms.

Robotic Car Synchronization (Embedded Systems)
Synchronized ESP32-based robotic cars using Python, I2C, sensors, and PID control achieving real-time coordination.

SKILLS

Programming Languages: C, C++, C#, Java, Python, JavaScript, Assembly x86
Frameworks & Tools: .NET Framework, Java Swing / JavaFX, MATLAB, ROS (Beginner), SFML, Arduino, ESP32
Systems & Networking: OS (Threads, Mutex, Semaphores), Cisco Packet Tracer, OSPF, EIGRP, RIP, NAT, ACLs, DHCP
Algorithms & Computing: Data Structures & Algorithms, Design & Analysis of Algorithms, Merkle Trees, Heaps, Treaps, Graph Algorithms, Numerical Computing, High Performance Computing (HPC), HPCG, SpMV
Databases & Tools: SQL Server, T-SQL, Schema Design, ETL, Git, Linux, Shell Scripting
Web & Other: HTML, CSS, JavaScript, Embedded Control Systems, Robotics Kinematics

Certifications & Courses

- **Robotics Technology Course,** FAST University Islamabad (2024): Learned forward/inverse kinematics and MATLAB simulations. - **Unity WorkShop** : Completed a 10-hour hands-on Unity workshop focused on core game development principles. - [Future Addition]: **Coursera Machine Learning** by Stanford University (In Progress).